

MLS Surface Reconstruction: Stationary-Ego Failure Case

Goal: reconstruct a continuous, sensor-agnostic static 3D scene from Waymo TOP LiDAR, then simulate a differently placed Valeo Scala-2 sensor.

Observed issue

A high-dynamic-object scene produced a visibly sparse MLS scene and fragmented Scala-2 raycast. Investigation showed that the problem is already present in the **pre-MLS accumulated ROAD observations**; it is not primarily caused by the raycaster.

Metric	Moving-ego scene	Stationary-ego / dynamic-rich
Frames	146	145
Ego trajectory	264.56 m	~0.01 m
Median ego step/frame	1.8070 m	0.0001 m
ROAD points	6,691,819	3,214,920
ROAD NN median	15.8 mm	0.3 mm
Occupied 10-cm XY cells	848,501	33,935
Occupied 20-cm XY cells	243,312	12,888

Interpretation

Moving ego: successive scans are acquired from different positions. Temporal accumulation therefore acts as spatial supersampling: the original LiDAR scan pattern shifts across the road and progressively fills the surface.

Stationary ego: the sensor repeatedly observes almost the same trajectories. Millions of points accumulate, but they are highly redundant and remain concentrated in narrow scan bands. This explains the extremely small nearest-neighbour distance together with ~19-25x fewer occupied spatial cells.

Why MLS struggles

Current ROAD MLS uses search radius 0.30 m, polynomial order 2, SAMPLE_LOCAL_PLANE upsampling radius 0.10 m, and step 0.075 m. This works when the input already provides broad local surface support. In the stationary scene, genuine unobserved strips remain between scan bands. Local MLS densification adds samples around existing measurements but does not perform general surface completion.

Observed Scala-2 consequence: rear-center frame 017, tangent-patch radius 0.05 m: **4,546 / 10,448 hits (43.51%)**, with visibly fragmented ground returns.

Role of dynamic objects

Dynamic objects likely compound the stationary-ego problem through occlusion. After removing a car or pedestrian from the static reconstruction, the ground hidden behind/below it may have no measurement. A moving ego can later observe such regions from another viewpoint; a stationary ego has much less opportunity to do so.

Potential research implication

This is a useful stress case for **sensor-agnostic reconstruction**. A method may appear successful when ego motion naturally fills the acquisition sensor's sampling gaps, yet fail when observations remain anisotropic and incomplete. For arbitrary novel LiDAR placement, the representation should model a continuous plausible surface rather than reproduce the original sensor's scan trajectories.

Core research question: How can we reconstruct continuous road/building surfaces from highly anisotropic, incomplete LiDAR observations without relying on ego-motion-induced densification?

Next controlled step

Quantify the cross-band/unobserved gap distribution in the stationary scene. Then test whether adaptive semantic MLS support can bridge realistic gaps without oversmoothing boundaries; if not, treat this as motivation for a stronger semantic/geometric surface-completion method.

Scenes: moving ego = segment-10203656353524179475_7625_000_7645_000; stationary/dynamic-rich = segment-17791493328130181905_1480_000_1500_000.